

1.0_Stage_1-ASK-Business_Questions

Stage 1: Ask - Key Business Questions

Questions likely to be asked by the company stakeholders from Animesh (Data Analyst)

1. Sales & Revenue Trends

1. **Monthly Growth:** What is the monthly revenue, and when do sales peak?
2. **Peak Traffic:** Which days and hours get the most orders?
3. **Average Spending:** What is the Average Order Value (AOV) per year?

2. Logistics & Delivery Performance

1. **Delivery Time:** What is the average delivery time, and which states have the slowest deliveries?
2. **Late Rates:** How many orders are delivered late?
3. **Shipping Costs:** Does high shipping cost cause order cancellations?

3. Customer Buying Behavior

1. **Payment Methods:** What are the most used payment methods, and do credit cards lead to higher spending?
2. **Installments:** Do payment installments increase order value?
3. **Top Spenders:** Which 10 cities have the highest-spending customers?

4. Product Catalog Insights

1. **Top Products:** Which product categories generate the most revenue and most sales?
2. **Late Penalty:** How much do late deliveries lower customer review scores?
3. **Listing Quality:** Do more photos and longer descriptions improve product review scores?

5. Predictive Analytics

1. **Predicting Satisfaction:** Can a Linear Regression model predict how much review scores drop per day of delay?

Questions to be answered by the Data Analyst to the company stakeholders in the project by adding Problem Definition & Key Analytical factors.

1. Sales & Revenue Trends (The Bottom Line)

1. **MoM Growth:**
What is the total revenue generated month-over-month, and are there distinct seasonal spikes (e.g., holiday seasons)?
2. **Peak Traffic:**
Which day of the week (purchase_day_of_week) and hour of the day (purchase_hour) see the highest volume of successful orders? (Actionable for timing marketing blasts or flash sales).

3. AOV (Average Order Value):

What is the average amount a customer spends per transaction, and how does this change over the years?

2. Logistics & Operational Efficiency

1. Delivery Baselines:

What is the average `delivery_time_days` across the platform, and which specific states (`customer_state`) suffer from the slowest average deliveries?

2. The Cost of Delay:

What percentage of total orders are flagged as `is_late_delivery`?

3. Freight Impact:

How does the `freight_ratio` (the cost of shipping relative to the item price) affect purchasing behavior? Do higher freight ratios lead to more canceled orders?

3. Customer Behavior & Segmentation

1. Payment Preferences:

What is the distribution of `payment_type`? Do customers using credit cards have a significantly higher AOV than those using other methods?

2. Installment Psychology:

Does the availability of `payment_installments` correlate with customers buying more expensive items (`total_item_value`)?

3. High-Value Customers (RFM Prep):

Which top 10 cities have the highest concentration of customers in the top 5% based on total payment value?

4. Product Catalog & Satisfaction

1. Volume vs. Value:

Which `product_category_name` drives the highest total revenue, and which drives the highest volume of individual items sold?

2. The Late Penalty:

Is there a direct, quantifiable drop in the average `review_score` when an order's `delivery_delay_days` is greater than 0?

3. Listing Quality:

Do products with longer descriptions (`product_description_lenght`) or more photos (`product_photos_qty`) statistically receive higher review scores or lower return rates?

5. Predictive Analytics & Future Impact

1. The Late Penalty Model:

Can we mathematically predict the exact drop in customer satisfaction for every additional day a delivery is delayed past its estimated arrival date using a Linear Regression model?



Author

[Animesh Sanghi](#) | Data Analyst | MBA '28 (Analytics & Data Science) MUJ

9406570600 | animeshsanghi.da@gmail.com | [LinkedIn Profile](#) | [GitHub Profile](#)